

Foldable Intraocular Lens (IOL) Implantation

Capsular Bag Refill, Controlled Unfolding, 360° Overlap & Viscoelastic Evacuation

OPTIC MATERIAL

Hydrophobic Acrylic (6.0 mm Optic)

HAPTIC DESIGN

Modified C-Loop (13.0 mm Total Length)

DELIVERY SYSTEM

Screw-Assisted Cartridge (2.4 mm Bore)

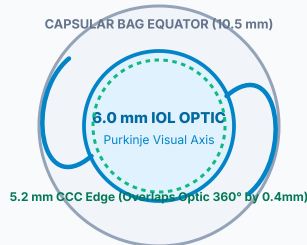
OPTIC EDGE

360° Posterior Square Edge

☀ What Is IOL Implantation in Simple Terms?

Once the cloudy cataract has been sucked out, the eye has no focusing lens—everything would be a blurry smudge! We replace the natural lens with a permanent, microscopic artificial lens made of foldable medical acrylic. Because our incision is only 2.4mm wide and the lens is 6mm wide, the lens is folded like a tiny taco inside a syringe. We inject it through the tiny wound, watch it unfold into the empty natural bag, center it precisely, and vacuum out the jelly.

IOL IN-THE-BAG ANATOMY, 360° RHEXIS OVERLAP & CENTRATION



WHY 360° OVERLAP MATTERS

1. Prevents PCO (Secondary Cataracts)

Square-edge optic + shrink-wrap overlap traps residual lens cells, stopping cloudy migration.

2. Perfect Centration & No Lens Tilt

Both spring arms seated symmetrically in bag prevents unwanted astigmatism and double vision.

STEP
1

Capsular Bag Refill with Cohesive OVD — Re-Inflating the Deflated Bag

Tool: Provisc Cohesive Jelly

☞ WHAT TO DO RIGHT NOW:

Take the Provisc cohesive viscoelastic syringe. Insert the cannula through the main wound and gently inject jelly directly into the equator of the empty lens bag until it puffs open into a firm, round pocket.

💡 **Why This Step Is Strictly Necessary:** Without the cataract inside, the thin natural bag collapses flat like an empty balloon. If you attempt to slide a hard plastic injector nozzle into a collapsed bag, the rigid tip will spear straight through the paper-thin back capsule! Injecting thick jelly acts like a balloon pump, expanding the bag in 3 dimensions and creating a safe landing pad for the new lens.

⚠ **Critical Hazard:** Attempting IOL delivery into a collapsed bag causes direct mechanical rupture of the posterior capsule.

STEP
2

Cartridge Delivery into Bag — Squeezing the Folded Lens Through the Keyhole

Tool: Monarch / Screw IOL
Injector

👉 WHAT TO DO RIGHT NOW:

Position the beveled tip of the injector nozzle into the 2.4mm clear corneal incision with the bevel pointing DOWN. Turn the screw plunger smoothly to advance the folded acrylic lens forward.

💡 **Why This Step Is Strictly Necessary:** The artificial lens is 6mm wide—more than double the size of our 2.4mm incision! The lubricated cartridge compresses the lens without tearing it. Advancing the plunger smoothly allows the lens to pass through the microscopic tunnel without stretching or burning the delicate cornea.

⚠️ **Critical Hazard:** Pushing too fast or unevenly can cause the spring-loaded lens to shoot forward with violent kinetic force, puncturing the posterior capsule.

STEP
3

Leading Haptic Placement — Guiding the Front Leg into the Corner

Tool: IOL Injector & Nozzle Guidance

👉 WHAT TO DO RIGHT NOW:

As the lens emerges from the nozzle, ensure the leading spring arm (haptic) uncurls directly into the far corner of the bag equator. Keep the nozzle tip inside the circular window so the lens doesn't escape over the iris.

💡 **Why This Step Is Strictly Necessary:** If the spring arms unfold on top of the colored iris instead of inside the bag, they will chafe against iris blood vessels, causing internal bleeding, painful inflammation, and chronic pupil distortion.

⚠️ **Critical Hazard:** Allowing the trailing arm to become trapped in the corneal tunnel can shear the haptic off the lens optic.

STEP
4

Sinskey Hook 360° Rotational Centering — Spinning & Tucking the Back Leg

Tool: 0.2mm Sinskey Micro-Hook

👉 WHAT TO DO RIGHT NOW:

Use the tiny angled Sinskey hook to engage the notch at the base of the trailing arm. Rotate the lens clockwise while gently pushing downward, tucking the trailing arm under the edge of the circular opening into the bag. Confirm 360° overlap.

💡 **Why This Step Is Strictly Necessary:** Both arms must be seated symmetrically inside the bag pockets. If one arm is in the bag and one is outside (asymmetric placement), the lens tilts, causing severe astigmatism, glare, and double vision. Having the circular capsular window overlap the 6.0mm lens edge all 360 degrees creates a biological seal that locks the lens in place permanently.

⚠️ **Critical Hazard:** Asymmetric haptic fixation triggers UGH Syndrome (Uveitis-Glaucoma-Hyphema) and chronic vision degradation.

STEP
5

Retro-Lens Viscoelastic Washout — Vacuuming Away the Dangerous Jelly

Tool: Coaxial I/A Handpiece

👉 WHAT TO DO RIGHT NOW:

Take the I/A suction wand. Gently tilt the new acrylic lens with the tip to slip into the pocket behind the lens. Thoroughly vacuum out all the thick cohesive jelly trapped in the back, then clean the front chamber.

💡 **Why This Step Is Strictly Necessary:** The surgical jelly that protected the eye earlier will cause severe damage if left behind! The microscopic drains of the eye (trabecular meshwork) get plugged up by thick jelly. Fluid cannot escape, causing intraocular pressure to soar past 45 or 50 mmHg within hours of surgery, causing excruciating eye pain, corneal edema, and irreversible blindness from optic nerve ischemia.

⚠️ **Critical Hazard:** Incomplete viscoelastic evacuation is the #1 cause of acute early post-operative glaucoma spikes (>40 mmHg).